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A Project Report on

Comparative Analysis of Machine Learning and Deep Learning for Skin Disease Detection

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CERTIFICATE

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It is hereby certified that the work which is being presented in the BTECH Major Project - III Report entitled **"Comparative Analysis of Machine Learning and Deep Learning for Skin Disease Detection"**, in partial fulfillment of the requirements for the award of the Bachelor of Technology in Electronics & Telecommunication Engg. and submitted to the **Department of Electronics & Telecommunication Engg.** of **MIT Academy of Engineering, Alandi(D), Pune, Affiliated to Savitribai Phule Pune University (SPPU), Pune**, is an authentic record of work carried out during Academic Year **2026–2027**, under the supervision of **Prof. Vinaya S. Tapkir**, **Department of Electronics & Telecommunication Engg.**

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DECLARATION

We the undersigned solemnly declare that the project report is based on our own work carried out during the course of our study under the supervision of **Prof. Vinaya S. Tapkir**.

We assert the statements made and conclusions drawn are an outcome of our project work. We further certify that

1. The work contained in the report is original and has been done by us under the general supervision of our supervisor.
2. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this Institute/University or any other Institute/University of India or abroad.
3. We have followed the guidelines provided by the Institute in writing the report.
4. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

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Foreword

Skin diseases are among the most common health issue's worldwide, often leading to discomfort, delayed treatment, and increased healthcare costs due to complex and time-consuming diagnostic procedures. Manual tests such as biopsy, dermoscopy, and laboratory tests require experts to analyze the report and require lot of time to process the tests which can lead to discomfort or risk to patient because of delay in treatment.

This project presents a comparative analysis of treditional ML models and deep learning methods like transformer for automatic skin disease classification using medical images. The proposed system uses advanced models including EfficientNet and Vision Transformer (ViT) to classify skin diseases into multiple categories. The dataset includes about 16,000 dermatological images across 23 classes. This project is based on multi class classification.

The methodology includes image pre-processing techniques such as resizing, normalization, and data augmentation esizing, Noise Reduction, Grayscale Conversion, Segmentation, Histogram Equalization (HE, AHE, CLAHE) at first for treditional ML model implementation by extracting image features using GLCM to feed to ML models.

Deep learning mpodels ike CNN ,MobilenetV2, efficientnet and Vision transformer is used for comparative analysis and feature extraction using these architectures to evaluate their performance in terms of accuracy, complexity, and training efficiency.

The results indicate that EfficientNet performed well on simpler datasets, while Vision Transformer achieved better performance on large datasets with high number

of classes by capturing global dependencies within images. The proposed system canbe used to identify the disease early, reducing the workload of healthcare professionals, and support telemedicine applications, thereby improving accessibility to dermatological care.

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Chapter 1

Introduction

1.1 Background

Skin diseases affect millions of people worldwide and range from relatively minor conditions such as acne and dermatitis to more serious diseases like melanoma and viral infections such as monkeypox (mpox). Accurate and timely diagnosis is essential for effective treatment and disease management (Kumar, Kumar, & Saboo, 2016). Traditionally, clinical analysis and laboratory tests such as biopsy, dermoscopy, patch testing, and microbial analysis, which are often time consuming and require expert analysis.

A skin biopsy removes a small sample of tissue for microscopic examination and mostly requires 3–7 days for diagnosis. Patch testing for allergic conditions takes 48–96 hours, while skin scraping and KOH tests for fungal infections may require several hours to a day. Similarly, viral PCR tests may take 1–3 days. These delays can impact treatment in time and disease control.

The recent global outbreak of mpox highlighted the limitations of conventional diagnostic methods, as it is not possible to detecct the small similarities in the diseases using human eye so it requires the laboratory tests. Early detection is critically important to improve the patient needs and prevent the spread of contagious diseases. Machine learning approaches have been used as trustable tools to enhance diagnostic accessibility and reduce clinician workload (Kumar et al., 2016; Srinivasu et al.,

2021).

Recent advances in Artificial Intelligence (AI) and Deep Learning (DL) have improved automated medical image analysis (Matsoukas et al., 2022). Convolutional Neural Networks (CNNs) have gave a best performance in image classification tasks (He, Zhang, Ren, & Sun, 2016; Srinivasu et al., 2021). Models such as EfficientNet (Tan & Le, 2019) include compound scaling and transfer learning to achieve high accuracy with low computation. Data augmentation techniques also improve models ability to generalize when labeled data is limited (Shorten & Khoshgoftaar, 2019).

In addition, transformer-based architectures are used for these tasks for better performance as it is able to capture global dependencies using self-attention mechanisms . The Vision Transformer (ViT) (Dosovitskiy et al., 2021) does the same . Further improvements such as DeiT (Touvron et al., 2021) and BEiT (Bao et al., 2022) enhance training efficiency and feature representation. The Swin Transformer (Liu et al., 2021) requires low computation using hierarchical structures.

Hybrid architectures combining CNN and transformer approaches, such as CvT (Wu et al., 2021) and advanced hybrid models (Li et al., 2025), capture both local and global features effectively. Lightweight models such as Edge-ViT (Pan et al., 2025) enable deployment on devices with limited resources.

Studies in medical imaging confirm the effectiveness of these models. Benchmark analyses (Chen et al., 2025) and surveys (Matsoukas et al., 2022) show that transformer-based models achieve best performance and can be effectively used with larger datasets.

In this project, a dataset containing around 16,000 dermatological images from 23 different skin disease categories is used to build a multi-class skin disease classification system. The proposed approach includes image preprocessing and data augmentation techniques (Shorten & Khoshgoftaar, 2019) to improve the quality and diversity of the dataset. A comparative study is also performed between CNN-based models (He et al., 2016; Tan & Le, 2019; Srinivasu et al., 2021) and transformer-based models (Dosovitskiy et al., 2021; Liu et al., 2021; Bao et al., 2022) to analyze their effectiveness in accurately classifying different skin diseases.

The rapid growth of artificial intelligence in the healthcare field shows that AI-based systems can provide quick, scalable, and cost-effective solutions for skin disease diagnosis. Such systems can support dermatologists by assisting in early and accurate detection while also making dermatological care more accessible to people, especially in areas where medical facilities are limited.

1.2 Project Idea

The main idea of this project is to design a diagnostic system that can classify skin diseases using deep learning techniques. The system takes a skin image as input and predicts the disease.

The project explores the use of two advanced models:

- Traditional ML models for classification
- Basic CNN and MobilenetV2 for small dataset
- EfficientNet – a convolutional neural network for feature extraction
- DINOv2 (Vision Transformer (ViT)) – A model that captures global image relationships

These models are used to compare their performance and choose best among them.

1.3 Motivation

Skin diseases can spread rapidly and may lead to dangerous complications in disease if not detected early. Recent outbreaks such as lumpy skin disease and monkeypox highlight the importance of timely diagnosis of the disease as these are very dangerous to life.

There is a significant gap between patients and dermatologists due to factors such as lack of awareness, high diagnostic costs, and the limited availability of specialists.

In many cases, manual diagnosis is time-consuming and the accuracy of the results may vary depending on the experience and expertise of the doctor.

These issues highlight the need for an automated system that can provide fast, accurate, and cost-effective diagnosis through image-based analysis. Deep learning techniques offer an efficient solution by improving diagnostic accuracy and assisting healthcare professionals in the early detection of skin diseases.

1.4 Project Challenges

The development of an automated skin disease classification system involves several challenges related to data quality, model performance, computational requirements, and the complexity of dermatological images. The major challenges encountered during this project are described below.

- **Dataset Diversity and Class Imbalance:** The DermNet dataset contains images belonging to multiple skin disease categories. The number of images varies across classes, which can cause the model to perform better on frequently represented classes and poorly on underrepresented classes.
- **Image Variability:** Skin images are different in terms of light, resolution,skin tone, lesion size. This can confuse model on unseen data.
- **Overfitting and Generalization:** Overfitting is problem where training accuracy is too high but and validation is too low. in this context model memorizes the training images too much that fails to recognize new images.
- **Computational Requirements:** for difffernet models, different resesourses and memory is required.
- **Preprocessing and Implementation:** For differnt architectures, different input diemntion is required. some DL models required normalization. data augmenatation technique should be used when there is data imbalance in dataset.
- **Reliable Disease Classification:** Some skin conditions look very same. Seperating them requires lot of similar data for model to understand small differences.

1.5 Proposed Solution

The proposed system consists of the following steps:

- **Data Collection:** Use of DermNet dataset containing approximately 16,000 images across 23 classes.
- **Pre-processing:** Image resizing, normalization, and augmentation, grayscale conversion techniques to improve model performance.
- **Model Training:** Train ML models, CNN, MobileNetV2, EfficientNet and DINOv2 Vision Transformer model.
- **Classification:** Multi-class classification of skin diseases.
- **Deployment:** Development of a user-friendly interface for image upload and prediction.

The system aims to improve diagnostic accuracy, reduce workload on healthcare professionals, and enable early detection of diseases.

1.6 Major Contribution

The key contributions of the project are as follows:

- **Multi-Class Skin Disease Classification:** Development of a classification pipeline using the DermNet dataset. This dataset contains approximately 16,000 dermatological images belonging to 23 skin disease categories.
- **Image Preprocessing and Augmentation:** Image should be resized to correct dimention as per requirement. As per our case data augmentation is required because dataset is imbalanced.
- **Comparative Model Analysis:** This projects contributes to Test different ML and DL models to compare accuracy.

- **Transfer Learning:** Use of pretrained CNN and Transformer Architectures can increase model capacity.
- **Potential Healthcare Support:** Through this system, Doctors can get help easilty to understand what exactly the disease is. It can help to reduce the long time tests and help decide the disease.

1.7 Project Report Organization

This report is organized into chapters to present the background, theoretical foundation, methodology, implementation, experimental results, and conclusions of the proposed skin disease classification system. The organization of the report is described below.

- **Chapter 1: Introduction** In introduction, Project motivation, its background. project idea, Proposed solution, challenges and major contribution is written.
- **Chapter 2: Literature Review** This literature review explores multiple papers related to skin disease classification which uses ML and DL techniques. These papers discusses multiple CNN, ML , Transformer architecture and find the challanges it found.
- **Chapter 3: Proposed Methodology** Describes the proper method implimented step by step.
- **Chapter 5: Results and Discussion** presents the results obtained from the implemented models.
- **Chapter 6: Conclusion and Future Scope** summarizes the project objectives, the major findings, and the conclusions drawn from the experimental evaluation. It also discusses the limitations of the proposed system and possible future improvements, such as using larger and more diverse datasets, improving minority-class recognition, and validating the models on independent clinical datasets.

The report concludes by presenting the references used throughout the study and any appendices containing supplementary implementation details or supporting material.

Chapter 2

Literature Review

The table below Table 2.1 represents a comparative analysis of 15 research studies related to image classification and skin disease detection. The selected papers cover different approaches like traditional machine learning techniques, convolutional neural networks (CNNs), and recent transformer-based architectures.

Early studies such as Kumar et al. (Kumar et al., 2016) focused on image processing combined with machine learning methods, which didn't give acceptable output but couldn't capture complex visual patterns. CNN-based models such as ResNet (He et al., 2016), EfficientNet (Tan & Le, 2019), and MobileNetV2 (Srinivasu et al., 2021) improved classification accuracy by extracting hierarchical features from images. Mostly in different models data augmentation techniques (Shorten & Khoshgoftaar, 2019) were widely used to enhance model generalization and reduce overfitting.

Recent research is using transformer-based models such as Vision Transformer (Dosovitskiy et al., 2021), DeiT (Touvron et al., 2021), and Swin Transformer (Liu et al., 2021), which uses self-attention mechanisms to capture global dependencies within images. Hybrid models like CvT (Wu et al., 2021) and CNN-transformer architectures (Li et al., 2025) improve performance by combining local and global features. Advanced models as BEiT (Bao et al., 2022) and training methods (Steiner et al., 2022) focus on improving efficiency and reducing dependency on large labeled datasets.

In the medical domain, studies such as Matsoukas et al. (Matsoukas et al., 2022) tells the effectiveness of transformer-based models in skin disease classification. Compar-

ative studies (Chen et al., 2025) and deployment based architectures such as Edge-ViT (Pan et al., 2025) get better results.

Table 2.1: Comparison of Existing Methods for Skin Disease and Image Classification

Ref	Model / Paper	Dataset	Technique	Accuracy	Advantages	Limitations
[1]	Kumar et al. (2016)	Dermatology Dataset	Machine Learning + Image Processing	Moderate	Simple implementation, low cost	Limited accuracy and scalability
[2]	Srinivasu et al. (2021)	HAM10000	MobileNetV2 + LSTM	∼85%	Fast and suitable for mobile devices	Confusion between similar classes
[3]	Matsoukas et al. (2022)	ISIC Dataset	ViT-based Survey for Medical Imaging	>90% AUC	High diagnostic performance	Requires large annotated datasets
[4]	He et al. (2016) – ResNet	ImageNet	Deep Convolutional Neural Network	High	Strong feature extraction capability	Lacks global contextual understanding
[5]	Tan & Le (2019) – EfficientNet	ImageNet / DermNet	CNN with Transfer Learning	∼85%	Efficient model scaling and performance	Limited global feature learning
[6]	Shorten & Khoshgoftaar (2019)	General Datasets	Data Augmentation Techniques	Improved	Enhances generalization ability	No standardized approach
[7]	Dosovitskiy et al. (2021) – ViT	ImageNet	Transformer-based Model	∼88%	Captures global dependencies effectively	Requires large-scale training data
[8]	Touvron et al. (2021) – DeiT	ImageNet	Data-efficient Transformer	∼82%	Performs well with limited data	Complex training process
[9]	Bao et al. (2022) – BEiT	ImageNet	Self-supervised Transformer	∼86%	Reduces dependency on labeled data	Computationally expensive
[10]	Liu et al. (2021) – Swin Transformer	ImageNet	Hierarchical Transformer	∼87%	Multi-scale feature learning	Architecturally complex
[11]	Wu et al. (2021) – CvT	ImageNet	CNN + Transformer Hybrid	∼87%	Combines local and global features	Increased parameter size
[12]	Li et al. (2025)	ImageNet	Hybrid CNN-Transformer Model	+2–3% improvement	Better generalization performance	Increased computational complexity
[13]	Pan et al. (2025) – Edge-ViT	Mobile / Edge Devices	Lightweight Transformer	∼78%	Suitable for edge deployment	Reduced accuracy vs. larger models
[14]	Chen et al. (2025)	Skin Disease Datasets	Comparative Benchmark Analysis	∼81%	Comprehensive evaluation	Performance depends on dataset
[15]	Steiner et al. (2022)	ImageNet	Training Optimization Techniques	Improved	Enhances ViT training efficiency	Sensitive to hyperparameter tuning
[16]	Oquab et al. (2023) – DINOv2	LVD-142M / ImageNet and other benchmarks	Self-supervised Vision Transformer	Benchmark-dependent; no single skin disease accuracy reported	Learns robust visual features with reduced dependence on labelled pretraining data	Computationally demanding; requires task-specific evaluation for skin disease classification

Chapter 3

Problem Definition and Scope

3.1 Problem Statement

Skin diseases are among the most common health conditions worldwide, affecting millions of individuals and causing symptoms such as rashes, inflammation, itching, and visible skin abnormalities. Early and accurate diagnosis is essential for effective treatment and prevention of disease progression. However, traditional methods such as biopsy, dermoscopy, and laboratory testing are really time consuming, expensive.

During earlier research, Many people have worked to make the systme skin disease detection automatic using image processing and machine learning techniques. For instance, Kumar et al. (Kumar et al., 2016) proposed a system based on basic image processing and classical machine learning algorithms, which provided moderate performance but couldent capture complex image features. Deep learning models like convolutional neural network (CNN) based models such as ResNet (He et al., 2016), EfficientNet (Tan & Le, 2019), and MobileNetV2 (Srinivasu et al., 2021) improved classification accuracy by learning hierarchical representations from images. Data augmentation techniques (Shorten & Khoshgoftaar, 2019) were used to enhance model generalization and reduce overfitting.

Even if use of CNN-based approaches cant capture much features and only focus on local feature extraction. To overcome these limitations, transformerb ased architectures such as Vision Transformer (ViT) (Dosovitskiy et al., 2021), DeiT (Touvron

et al., 2021), and Swin Transformer (Liu et al., 2021) were used by many professionals. These models use self-attention mechanisms to capture global dependencies, which improves model performance on complex datasets. Hybrid approaches such as CvT (Wu et al., 2021) and CNN-transformer models (Li et al., 2025) combine local and global feature extraction to enhance classification accuracy.

In many regions such as rural and under resourced areas, dermatologists are not present, which leads to lot of travel to reach to city and delay in diagnosis and improper treatment. Additionally, manual diagnosis requires experts analysis and require lot of time to process, which may result in inconsistent outcomes.

Therefore, there is a need to develop an automated, and efficient skin disease classification system that uses CNN-based and transformer-based techniques.

3.2 Goals and Objectives

The main goal of this project is to design and develop an skin disease detection system using deep learning techniques.

The specific objectives are as follows:

- To collect and preprocess a dermatological image dataset consisting of approximately 16,000 images across 23 skin disease classes.
- to compare different Classical ML models with their performance.
- To implement deep learning models such as EfficientNet (CNN-based) and Vision Transformer (ViT).
- To compare performance of CNN-based and transformer-based models.
- To enhance model performance using pre-processing and data augmentation techniques (Shorten & Khoshgoftaar, 2019).
- To develop a API for real-time skin disease prediction.
- To create a responsive chatbot for further questioning related to result

3.3 Scope and Major Constraints

Scope:

- The system focuses on multi-class classification of skin diseases using image inputs.
- It can assist healthcare professionals in early diagnosis and decision making.

Major Constraints:

- Availability and quality of labeled dermatological datasets.
- Class imbalance across different disease categories.
- High computational requirements for training transformer based models such as ViT.
- Requirement of large datasets for better generalization and performance.
- Limited real-world validation and clinical testing.

3.4 Hardware and Software Requirements

Hardware Requirements:

- System with minimum 8GB RAM (16GB recommended)
- GPU
- Storage: Minimum 20GB

Software Requirements:

- Programming Language: Python
- Frameworks: TensorFlow / PyTorch
- Libraries: NumPy, OpenCV, Matplotlib, Scikit-learn
- Development Tools: Jupyter Notebook / VS Code

3.5 Expected Outcomes

- A trained deep learning model capable of accurately classifying multiple skin diseases.
- Comparative performance evaluation of different ML and DL models
- A user-friendly API for uploading images and getting predictions in percentages.

Chapter 4

System Requirement Specification

4.1 Overall Description

The proposed system takes the input images through API and predicts disease and displays the 3 diseases with highest probability. The system uses the DINOv2 to achieve high accuracy in the classification multiclass image data set.

The system takes an input image, does pre-processing, extracts features using deep learning models, and outputs the predicted disease class.

4.1.1 Use Case Description

- User uploads a skin image
- System pre-processes the image
- Model analyzes the image
- System displays predicted disease

4.2 Data Sets Decsription

1. **DermNet Dataset:**

   The dataset used in this project consists of approximately 16,000 dermatological images collected kaggle dataset. It is designed for multi-class skin disease classification using deep learning techniques.

   The dataset includes both diseased and normal skin images. It contains a total of 23 classes representing different types of skin diseases.

   Dataset Characteristics:

   - Total Images: ∼16,000
   - Number of Classes: 23
   - Image Type: RGB images
   - Problem Type: Multi-class classification

   This dataset is primarily used for training the deep learning models. As it has diverse diseases and multiple classes.

2. **MSID Dataset (Monkeypox Skin Image Dataset):**

   - Contains 770 images across 4 classes: monkeypox, chickenpox, measles, and normal skin.

4.3 Mathematical Modelling

The primary image-classification experiment uses the DermNet collection described in this work, comprising approximately 16,000 images across 23 disease categories. A separate experiment uses the Monkeypox Skin Image Dataset (MSID), described in the project as containing 770 images from four categories: chickenpox, measles, monkeypox, and normal skin. These datasets serve different evaluation settings and their results are reported separately; the four-class MSID results are not directly interchangeable with the 23-class DermNet results.

Let the labelled dataset be

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}, \tag{4.1}$$

where x_i represents an input and $y_i \in \{1, \ldots, C\}$ represents correct output, N means no. of input images while C shows no. of classes($C = 23$ for DermNet and $C = 4$ for the stated MSID experiment). Each image must retain its correct label throughout preprocessing. Training, validation, and test partitions should be separated before augmentation so that transformed copies of a test image cannot enter the training data.

4.3.1 Image Pre-processing and Data Augmentation

Dataset is resized to 224×224 pixels to provide a consistent data shape for the selected networks. Images are then converted to similar range by using preprocessing. The normalization operation can be expressed generally as

$$x'_{i,j,c} = \frac{x_{i,j,c} - \mu_c}{\sigma_c}, \tag{4.2}$$

where $x_{i,j,c}$ is the color value at a specific pixel spot (i, j) for channel c and μ_c and σ_c are the average value and the variation spread of that entire color channel. The exact normalization constants must match the selected model's preprocessing implementation; Eq. (4.2) describes the general operation rather than asserting a particular set of constants.

To reduce overfitting and improve robustness to image orientation and framing, training images may be augmented using rotations, horizontal/vertical flips, small trans-

lations, and zoom transformations. An augmentation can be represented as

$$\tilde{x}_i = T(x_i; \theta), \tag{4.3}$$

where T is a transformation and θ denotes its sampled parameters. Training images are balanced by Flipping, rotationg, cropping the images randomply. Contrast Limited Adaptive Histogram Equalization (CLAHE) is an optional contrast enhancement step for images in which local contrast is insufficient. Since contrast enhancement can also alter diagnostically relevant appearance, it should be applied consistently and its use documented for the experiment in which it is enabled.

4.3.2 ML and DL Model Comparison

ML and DL use different techniques to detect skin disease images. Traditional machine learning models operate on feature vectors, whereas DL models learn image contents and learn to seperate important features from image. The mathematical formulations of the evaluated models are summarized below.

For traditional machine learning models, let $\mathbf{f}_i \in \mathbb{R}^d$ represent the feature vector associated with image x_i.

For Support Vector Machine (SVM), the binary decision function is

$$f(\mathbf{f}_i) = \mathbf{w}^T \mathbf{f}_i + b, \tag{4.4}$$

here $\mathbf{w}$ represents the weight of features and b shows bias of output. The soft-margin optimization objective is

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C_{\text{svm}} \sum_{i=1}^{N} \xi_i, \tag{4.5}$$

is subjected to

$$y_i(\mathbf{w}^T \mathbf{f}_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0, \tag{4.6}$$

where $y_i \in \{-1, +1\}$, ξ_i represents slack, and C_{svm} controls penalty for classification errors. Multi-class classification can be performed using one-versus-rest or one-versus-one strategies.

Random Forest combines the predictions of multiple decision trees. Its predicted class is determined by majority voting:

$$\hat{y}_i = \text{mode}\{T_1(\mathbf{f}_i), \ldots, T_M(\mathbf{f}_i)\}, \tag{4.7}$$

where T_m represents prediction of tree m whereas M represents exact no. of trees.

XGBoost constructs an additive ensemble of decision trees. At different iteration at t, output updates at

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(\mathbf{f}_i), \tag{4.8}$$

where f_t represnts new tree. Its regularized objective is

$$\mathcal{L}^{(t)} = \sum_{i=1}^{N} l(y_i, \hat{y}_i^{(t)}) + \sum_{t'=1}^{t} \Omega(f_{t'}), \tag{4.9}$$

where l menas classification loss. Ω penalizes model complexity.

CatBoost also uses an additive tree ensemble:

$$F_T(\mathbf{f}_i) = \sum_{t=1}^{T} f_t(\mathbf{f}_i), \tag{4.10}$$

where f_t represents the tree added at iteration t. For multi-class classification, the ensemble produces one logit for each class, which can be converted to probabilities using the softmax function in Eq. (4.20).

For deep learning models, the image representation is learned through trainable neural network layers. MobileNetV2 splits normal 3D convolution in 2 steps to reduce computational workload. The approximate cost of a standard convolution is

$$\text{Cost}_{\text{standard}} = HWK^2C_{\text{in}}C_{\text{out}}, \tag{4.11}$$

H and W both are the map of the features, K represents the kernel size. C_{in} and C_{out} are in/out channel counts. Corresponding approximate workload of a depthwise convolution then pointwise convolution is

$$\text{Cost}_{\text{separable}} = HW(K^2C_{\text{in}} + C_{\text{in}}C_{\text{out}}). \tag{4.12}$$

EfficientNet adjusts the range of depth, resolution, width using the compound scaling principle given in Eq. (4.13) and Eq. (4.14). DINOv2 extracts contextual image features using transformer self-attention, as formulated in Eq. (4.17) and Eq. (4.18). The extracted representation is mapped to class logits using Eq. (4.19).

4.3.3 EfficientNet Based Classification

EfficientNet is baseline of CNN. It's design adjusts the range of depth, width, and resolution to maintain coordination. CNN blocks learn local visual features which are lesion boundaries, chnages in colors and texture. In transfer learning, a pretrained model is connected to a task specific classification head with C output units. The final layer produces logits $\mathbf{z} = [z_1, \ldots, z_C]$ for the target classes.

EfficientNet's compound scaling principle is commonly expressed as

$$d = \alpha^{\phi}, \qquad w = \beta^{\phi}, \qquad r = \gamma^{\phi}, \tag{4.13}$$

under the condition that the approximate resource constraint

$$\alpha\beta^2\gamma^2 \approx 3, \qquad \alpha \geq 1,\ \beta \geq 1,\ \gamma \geq 1, \tag{4.14}$$

where d is depth, w is width, and r is resolution scaling factors, ϕ changes model scale, and α, β, and γ are constants selected by the architecture design. These equations describe the EfficientNet design principle; they do not mean that these coefficients were separately tuned in the present experiments.

4.3.4 DINOv2-Based Classification

DINOv2 (Oquab et al., 2023) is a vision model that learns general-purpose visual representations from images without requiring manually assigned class labels during its original pretraining. It is Vision Transformer architecture, uses self-attention to capture relationships between different image regions. In this work, DINOv2 is considered for dermatological image classification by connecting its learned visual representation to a task-specific classification head.

for $H \times W$ with C_{in} channels patch size $P \times P$, the number of non-overlapping patches is

$$N_p = \frac{HW}{P^2}, \tag{4.15}$$

considering that H and W can be devided P. Each patch is flattened to a D-dimensional embedding:

$$\mathbf{e}_j = \mathbf{x}_p^j \mathbf{E}, \qquad j = 1, \ldots, N_p, \tag{4.16}$$

where $\mathbf{x}_p^j$ is the flattened representation of patch j and $\mathbf{E}$ is the patch-projection matrix. The resulting patch embeddings are processed by the transformer encoder along with the positional information and special tokens used by the selected DINOv2 architecture.

For Q, K, and V, scaled dot product attention is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{4.17}$$

d_k shows key vector. The scaling term $\sqrt{d_k}$ helps to prevent big dot products from pushing the softmax into regions with smallest gradients. Multi-head attention applies this operation to multiple learned parameters.

Let the pretrained (?, ?) DINOv2 feature extractor be represented by F_θ. The feature map of image x_i is

$$\mathbf{h}_i = F_\theta(x_i), \tag{4.18}$$

where $\mathbf{h}_i \in \mathbb{R}^{D_f}$ is the image-level vector which shows features and D_f is feature dimension to the selected model. The task-specific classification head maps this representation to C class logits:

$$\mathbf{z}_i = W_{\text{cls}}\mathbf{h}_i + \mathbf{b}_{\text{cls}}, \tag{4.19}$$

where $W_{\text{cls}} \in \mathbb{R}^{C\times D_f}$ is classifier weight matrix & $\mathbf{b}_{\text{cls}} \in \mathbb{R}^C$ shows the bias vector. The logits are converted to probabilities using Eq. (4.20), and predicted class is obtained using Eq. (4.21).

Depending on the implementation, the pretrained DINOv2 backbone may be frozen while the classification head is trained, or the backbone and classification head may be fine-tuned jointly. The selected strategy must match the actual training configuration.

4.3.5 Output Probabilities and Training Objective

For either model, the output logits are converted into probabilities by the softmax function:

$$p_{i,c} = \frac{\exp(z_{i,c})}{\sum_{k=1}^{C}\exp(z_{i,k})}, \tag{4.20}$$

where $z_{i,c}$ is the logit for class c for image i. The predicted class is

$$\hat{y}_i = \arg \max_{c \in \{1,\ldots,C\}} p_{i,c}. \tag{4.21}$$

For classificationof single lable of multiple classes, categorical cross-entropy is a suitable objective:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{i=1}^{B} \sum_{c=1}^{C} y_{i,c} \log(p_{i,c} + \epsilon), \tag{4.22}$$

where B indicates the batch size, $y_{i,c}$ indicates the one-hot ground truth indicator, $p_{i,c}$ is the probability of prediction , and ϵ is a small numerical constant. The loss penalizes low probability assigned to the correct class. Model parameters are updated by gradient-based optimization; the project implementation describes use of Adam. The learning rate, batch size, and epoch count must be reported per run from the actual training configuration rather than inferred from the ranges in a draft.

4.3.6 Evaluation Metrics

As per accuracy calculates the corrects images per all test images:

$$\text{accuracy} = \frac{\sum_{i=1}^{N_{\text{test}}} \mathbb{1}(\hat{y}_i = y_i)}{N_{\text{test}}} \times 100\%, \tag{4.23}$$

where $\mathbb{1}(\cdot)$ is 1 when its argument is true and 0 otherwise. For an individual class c, the table reports the proportion of that class's test images that were correctly classified:

$$\text{ClassAccuracy}_c = \frac{\text{Correct}_c}{\text{Test}_c} \times 100\%. \tag{4.24}$$

accuracy is calculated only by taking correct predictions from all predictions which are related to per class recall, not the one vs rest accuracy that includes true negatives. Precision and recall for class c can be calculated using one vs rest counts:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \tag{4.25}$$

and the class wise F1-score is

$$F1_c = 2\frac{\text{PrecisionRecall}}{\text{Precision} + \text{Recall}}. \tag{4.26}$$

A confusion matrix shows count of true labels against predicted labels. For multiclass evaluation, evalution metrics shown above are preferable to relying on accuracy alone, particularly when class distributions differ.

4.4 Project Planning

- Dataset collection
- Data pre-processing
- Model development (EfficientNet, ViT)
- Training and evaluation
- Deployment

Chapter 5

Proposed Methodology

5.0.1 ML based proposed method

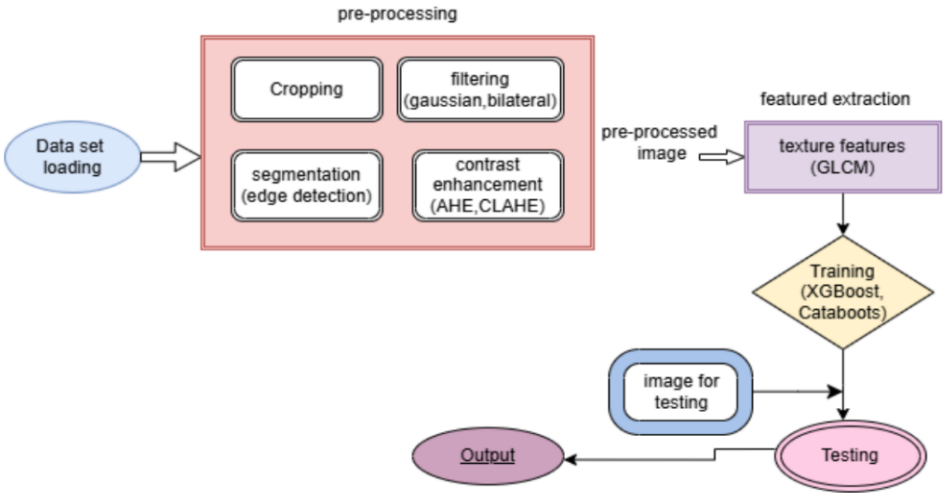


Figure 5.1: ML based proposed method

Description of Proposed System:

Above figure 5.1 shows complete image classification pipeline designed for skin disease detection (Kumar et al., 2016). in this process first dataset is loaded the preprocessed and features are extracted from images and fed to Different m ML models to get prediction.

contrast enhancement using methods like Adaptive Histogram Equalization (AHE) and Contrast Limited AHE (CLAHE) (Shorten & Khoshgoftaar, 2019) to improve image visibility and detail.

Once pre-processing is complete, the images undergo feature extraction, where texture features are computed using the Gray-Level Co-occurrence Matrix (GLCM) (Kumar et al., 2016).

1. The extracted features are fed to ML models to train ML Models.
2. During testing test set is given to model to checck model performance
3. This diagram represents a machine learning pipeline for skin disease detection using image data.
4. It includes dataset loading, preprocessing, segmentation, and contrast enhancement.
5. GLCM method is applied for feature extraction.
6. traditional Machine learning techniques are used for classification.

1] Dataset Loading: Datasets like MSID and DermNet provide diverse, high-quality dermatological images for training (Kumar et al., 2016).

2] Image Pre-Processing:

- Resizing
- Normalization
- Data Augmentation (Shorten & Khoshgoftaar, 2019)
- Noise Reduction
- Grayscale Conversion
- Segmentation
- Histogram Equalization (HE, AHE, CLAHE)

1. **Resizing:** Ensures input size is in correct size.
2. **Normalization:** Scales pixel values.
3. **Data Augmentation:** Improves generalization (Shorten & Khoshgoftaar, 2019).
4. **Noise Reduction:** Removes noise from dataset.
5. **Grayscale Conversion:** Reduces complexity.
6. **Segmentation:** Identifies regions of interest.
7. **Histogram Equalization:** Enhances contrast.

3] Feature Extraction (GLCM):

GLCM captures spatial relationships between pixels and extracts texture features such as contrast, correlation, energy, homogeneity, and entropy (Kumar et al., 2016).

GLCM Feature Extraction Process:

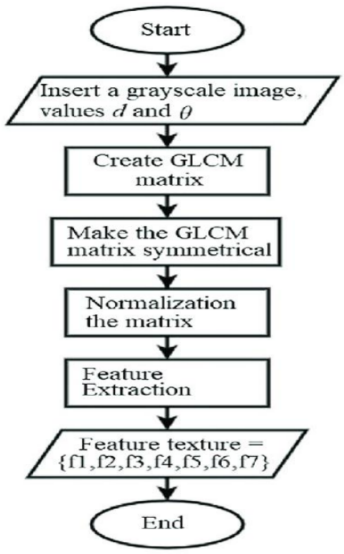


Figure 5.2: GLCM Flowchart

As shown in figure 5.2, this is a feature extraction process.

1. **Start:** The process begins.
2. **Insert a grayscale image, values d and θ:** A grayscale image is input along with two parameters — d (distance between pixel pairs) and θ (orientation angle, e.g., 0°, 45°, 90°, 135°) — which define the spatial relationship of pixels for GLCM computation.
3. **Create GLCM matrix:** The GLCM is generated based on the spatial relationship defined by d and θ. It records how often pairs of pixel intensities occur in the specified configuration.
4. **Make the GLCM matrix symmetrical:** The matrix is made symmetrical, typically by adding it to its transpose, so that pixel relationships in both directions are considered.
5. **Normalization of the matrix:** The matrix is normalized so that its elements represent probabilities (each value is divided by the sum of all matrix elements).
6. **Feature Extraction:** Many features like contrast, correlation, energy, and homogeneity are extracted from scaled GLCM,
7. **Feature texture = {f1, f2, ..., f7}:** A set of texture features (f1–f7) is obtained. These may include:
    - Contrast
    - Correlation
    - Energy
    - Homogeneity
    - Dissimilarity
    - ASM (Angular Second Moment)
    - Entropy
8. **End:**

4]Training (Classification and Regression)

Extracted features are used to train models like:

XGBoost: XGBoost, which stands for Extreme Gradient Boosting, is a machine learning algorithm based on the gradient boosting technique (Kumar et al., 2016). In this approach, decision trees are created sequentially, where each new tree tries to reduce the errors made by the previous trees. The model improves its performance step by step using gradient descent optimization, which helps achieve better prediction accuracy.

XGBoost can be used for both classification and regression tasks and is widely known for its high speed and strong performance. One of its important advantages is the use of built-in regularization techniques such as L1 and L2 regularization, which help reduce overfitting and improve the model's ability to generalize on unseen data. The algorithm is also capable of handling missing values automatically and supports parallel processing, making the training process faster and more efficient.

In addition, XGBoost uses tree pruning methods to avoid unnecessary complexity in the model and also provides support for custom objective functions. Due to its accuracy, efficiency, and ability to handle noisy datasets effectively, XGBoost has become one of the most widely used machine learning algorithms in both research and real-world applications.

- It builds an ensemble of decision trees.
- These trees correct errors from previous trees
- Handles missing data and noisy inputs.

CatBoost: Handles categorical data and reduces overfitting. CatBoost, short for Categorical Boosting, is a gradient boosting algorithm developed by Yandex, It does not require one ot encoding or label encoding. It uses ordered boosting, which prevents overfitting by avoiding target leakage. CatBoost includes symmetric trees as base learners, improving training efficiency. It is really good for imbalanced datasets,

supports GPU acceleration, handles missing values, and includes built-in hyperparameter optimization tools.

- Designed to handle categorical data without any encoding.
- Uses ordered boosting to prevent overfitting.
- Provides faster training and better performance on imbalanced data.

SVM: Finds optimal separating hyperplane for classification (He et al., 2016). Support Vector Machine (SVM) is used for both classification and regression tasks.SVM finds the optimal hyperplane which separates data classes while maximizing the margin between the closest data points to margin called support vectors.

5.1 DL based proposed method system

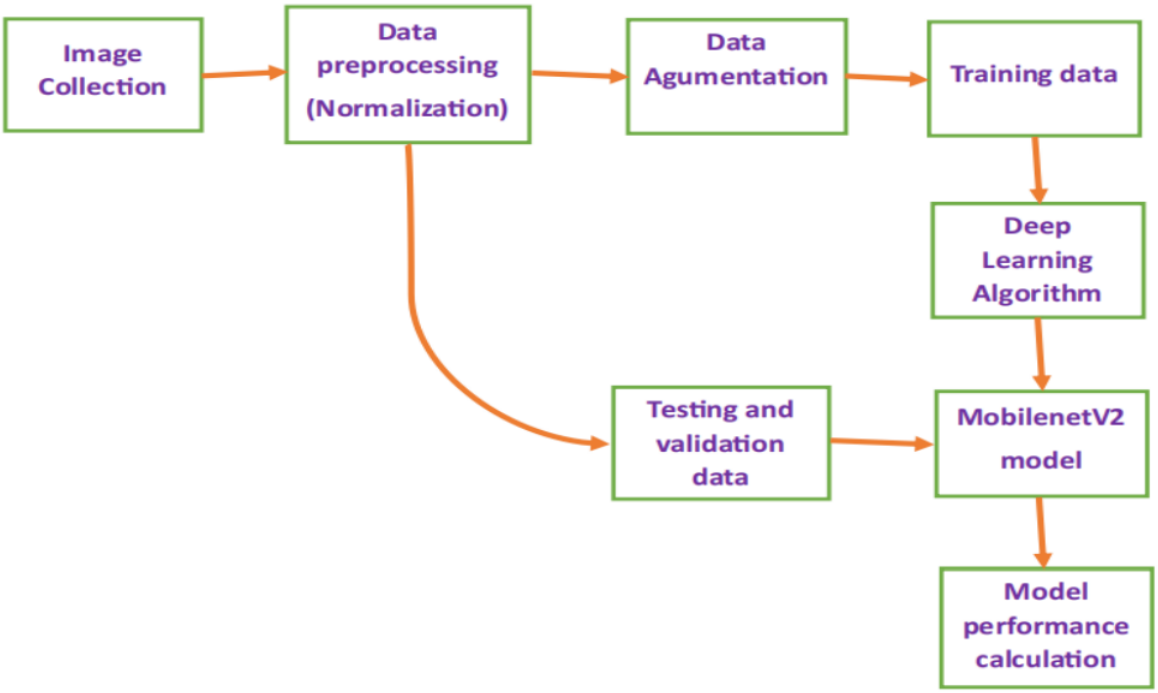


Figure 5.3: DL based proposed method system

The deep learning pipeline includes preprocessing, augmentation, and CNN-based classification.

- Images resized to 224×224
- Normalization applied
- Augmentation improves generalization (Shorten & Khoshgoftaar, 2019)

MobileNetV2 Model:

MobileNetV2 is a lightweight CNN architecture suitable for real-time applications and medical imaging tasks (Srinivasu et al., 2021).

The model is evaluated using accuracy, precision, recall, and F1-score.

Deployment is achieved using API-based integration for real-time predictions.

5.2 EfficientNet and Vision Transformer model

To improve classification performance, advanced architectures such as EfficientNet and Vision Transformer (ViT) are used.

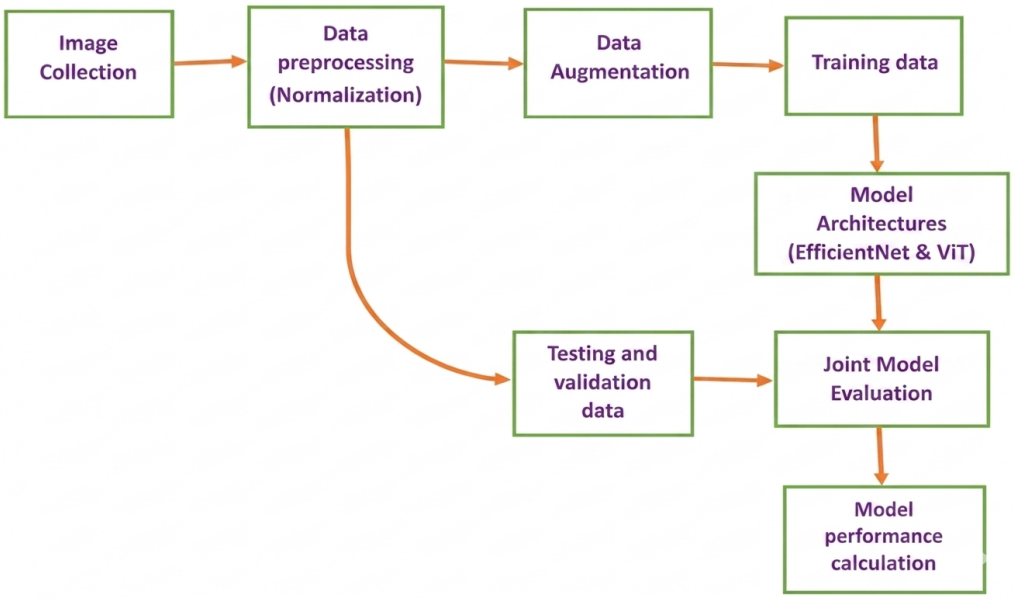


Figure 5.4: Proposed System Architecture using EfficientNet and Vision Transformer

EfficientNet

EfficientNet uses compound scaling to balance depth, width, and resolution efficiently (Tan & Le, 2019). It improves feature extraction while giving low computation requirement.

$$F(i,j) = \sum_{m} \sum_{n} I(i+m, j+n) \cdot K(m,n)$$

Working of EfficientNet: EfficientNet is a Convolutional Neural Network (CNN) designed to improve performance by scaling three important dimensions of the network: depth (number of layers), width (number of channels), and resolution (input image size) (Tan & Le, 2019). Instead of increasing these parameters randomly, EfficientNet uses a compound scaling method that balances all three dimensions efficiently.

The working of EfficientNet can be explained in the following steps:

- **Input Processing:** The input image is resized and normalized.
- **Convolution Operations:** The image passes through multiple convolution layers where uses filters (kernels) to extract local features such as edges, textures, and color patterns. This is mathematically represented as:

$$F(i,j) = \sum_m \sum_n I(i+m, j+n) \cdot K(m,n)$$

- **Feature Hierarchy:** Initial layers capture low level features (edges, corners), while deeper layers capture high level features (shapes, lesions).
- **Mobile Inverted Bottleneck (MBConv):** EfficientNet uses MBConv blocks which reduce computation while maintaining accuracy by expanding and compressing feature maps efficiently.
- **Global Feature Extraction:** Final layers combines features using pooling operations.
- **Classification:** Fully connected layers with Softmax produce the final class probabilities.

Thus, EfficientNet focuses on extracting detailed **local spatial features**, which are crucial for identifying fine patterns in skin diseases.

DINOv2

DINOv2 is a self-supervised vision foundation model that learns robust visual features from images without requiring manual labels during pretraining (Oquab et al., 2023). It is based on the Vision Transformer (ViT) architecture and uses self-attention to capture relationships between different image patches.

The working of DINOv2 is as follows:

- **Image Patching:** The input image is divided into small fixed size patches. Each patch is treated as a token for further processing.

- **Patch Embedding:** Each patch is converted into a vector and projected into a higher dimensional feature representation.
- **Self-Supervised Pretraining:** DINOv2 learns meaningful visual representations from large collections of unlabeled images, reducing its dependence on manually annotated training data.
- **Self-Attention Mechanism:** The model applies self-attention to learn relationships between different image patches:

  $$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

  This enables the model to capture relationships between local image regions and learn global contextual information.
- **Visual Feature Extraction:** The transformer encoder generates rich visual representations containing information about patterns, textures, shapes, and other important image characteristics.
- **Fine-Tuning:** The pretrained DINOv2 model can be fine-tuned using the labelled skin disease dataset to adapt its learned representations to the target classification task.
- **Classification Head:** The extracted features are passed to a classification layer, which predicts the corresponding skin disease category.

As a result, DINOv2 learns robust visual representations that can be adapted to skin disease classification. Its self-supervised pretraining enables the model to learn useful image features before being trained on the target dataset. These features can help distinguish visual patterns associated with different skin conditions (Oquab et al., 2023; Matsoukas et al., 2022; Chen et al., 2025).

The use of pretrained DINOv2 features and fine-tuning provides a foundation for improving skin disease classification. The final performance depends on the quality and diversity of the dataset, the fine-tuning strategy, and the evaluation procedure.

Chapter 6

Implementation

6.1 System Implementation

The proposed system is implemented using deep learning techniques for automatic skin disease classification. The system is developed in Python using frameworks such as TensorFlow and PyTorch. During implementation multiple stages including data preprocessing, model training, and prediction were performed.

The DermNet dataset consisting of approximately 16,000 images across 23 classes is used. The dataset includes both normal and diseased skin images. The system processes these images and trains deep learning models such as EfficientNet and Vision Transformer (ViT).

6.2 Implementation Parameters

The following parameters are used during model training:

- Image Size: 224×224 pixels
- Batch Size: 16 / 32
- Learning Rate: 0.001 (depending on the model)
- Optimizer: Adam for deep learning models
- Loss Function: Categorical Cross-Entropy for multi-class classification
- Number of Epochs: 20–50 (depending on the model)

For machine learning models, extracted features are used as input to classifiers such as SVM, XGBoost, and CatBoost. For deep learning models, image preprocessing and data augmentation techniques such as rotation, flipping, and zooming are used to improve model generalization and reduce overfitting. DINOv2 uses pretrained visual representations, which are adapted to the target dataset through fine-tuning.

Data augmentation techniques such as rotation, flipping, and zooming are used to improve model generalization and reduce overfitting.

6.3 User Interface

The system provides a simple and user-friendly interface that allows users to upload skin images and obtain predictions.

- User uploads a skin image through the interface
- The system preprocesses the image automatically
- The trained model analyzes the image
- The predicted disease class is displayed as output

6.4 Data Description

The dataset used in this project is the DermNet dataset.

- Total Images: Approximately 16,000
- Number of Classes: 23
- Image Type: RGB images
- Problem Type: Multi-class classification

The dataset contains images of various skin conditions which gives model variations in data to learn about multiple diseases.

6.5 Functional Implementation

The system performs the following functions:

- **Image Collection:** images are collected from the dataset or uploaded by the user.
- **Pre-processing:** Image resizing, normalization, and, where applicable, data augmentation are performed.
- **Feature Extraction:** Traditional machine learning uses image features such as Gray-Level Co-occurrence Matrix (GLCM) features. Deep learning models such as MobileNetV2 and EfficientNet learn features automatically. DINOv2 extracts pretrained visual representations.
- **Classification:** SVM, XGBoost, and CatBoost are used for machine learning classification, while deep learning models use a classification layer to predict the disease category.
- **Model Evaluation:** Model performance is evaluated using suitable metrics, including accuracy, precision, recall, F1-score, and confusion matrix.

- **Model Saving:** The trained models and required preprocessing configurations are saved and can be reused for prediction without retraining.

The implementation enables comparison between traditional machine learning, conventional deep learning, and the DINOv2-based approach for skin disease classification.

The trained model is saved and can be reused for prediction without retraining.

Chapter 7

Result

7.1 ML based proposed method system



Figure 7.1: Dataset Load and resize

The above figure 7.1 displays the output of that processes a dermatological image dataset for use in a medical image classification. The log messages displayed indicate that a total of 20,329 images were successfully identified and resized, with the process reaching 100% completion. Only random 5 images are displayed from dataset.

7.2 Applied CLAHE

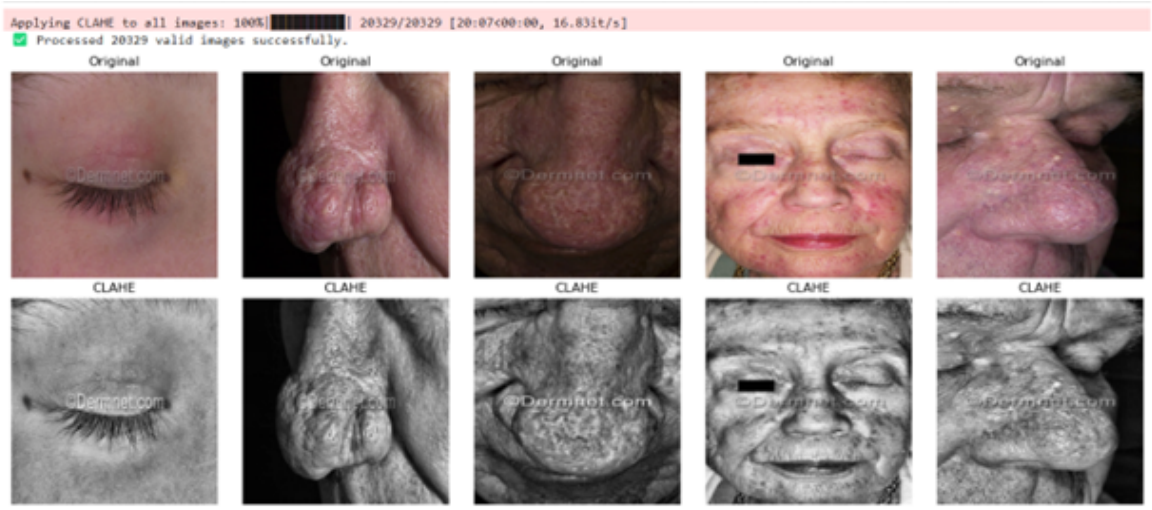


Figure 7.2: Apply CLAHE

Above figure 7.2 shows the result of applying a medical image preprocessing technique called CLAHE (Contrast Limited Adaptive Histogram Equalization) to a dataset of dermatological images. The purpose of CLAHE is to enhance contrast in images. CLAHE highlights features like skin texture, lesions, and other abnormalities.

7.3 GLCM graph



Figure 7.3: GLCM graph

As shown in the figure 7.3 the graph shows the variation of GLCM (Gray-Level Co-occurrence Matrix) texture features across the first 50 images in dataset. Each line represents a different texture feature, including contrast, dissimilarity, homogeneity, energy, correlation, and ASM (Angular Second Moment). Among these, contrast (shown in red) shows the most variation in data, with values ranging from around 200 to over 1400. Contrast is dominant feature among all.

7.4 ML models applied

7.4.1 XGboost

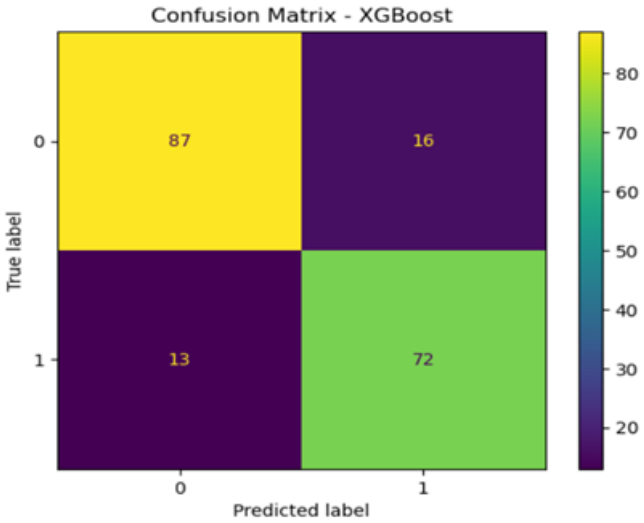


Figure 7.4: XGboost confusion matrix

The confusion matrix in Figure 7.4 shows that the XGBoost model correctly predicted 87 samples as class 0 and 72 samples as class 1. The model made 16 false positive predictions and 13 false negative predictions. The overall accuracy achieved by the model is approximately 84.68%.

7.4.2 CatBoost

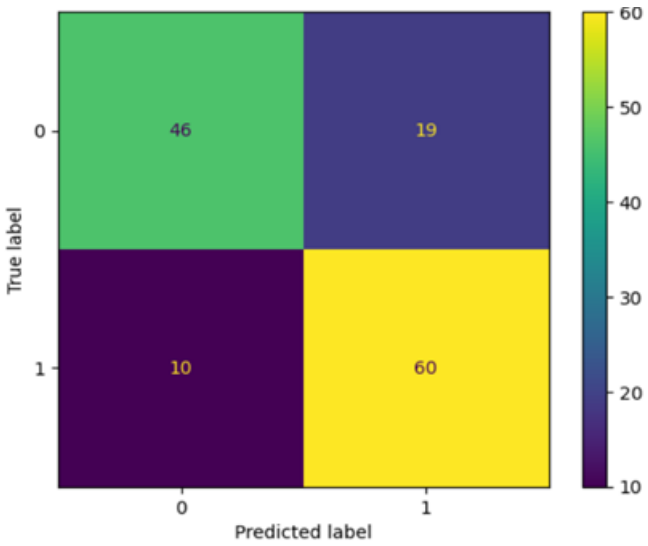


Figure 7.5: Catboost confusion matrix

The confusion matrix in Figure 7.5 shows that the model correctly predicted 46 samples as class 0 and 60 samples as class 1. It also produced 19 false positive predictions and 10 false negative predictions. The model achieved an overall accuracy of 78.52%, indicating reasonably good classification performance.

7.4.3 SVM

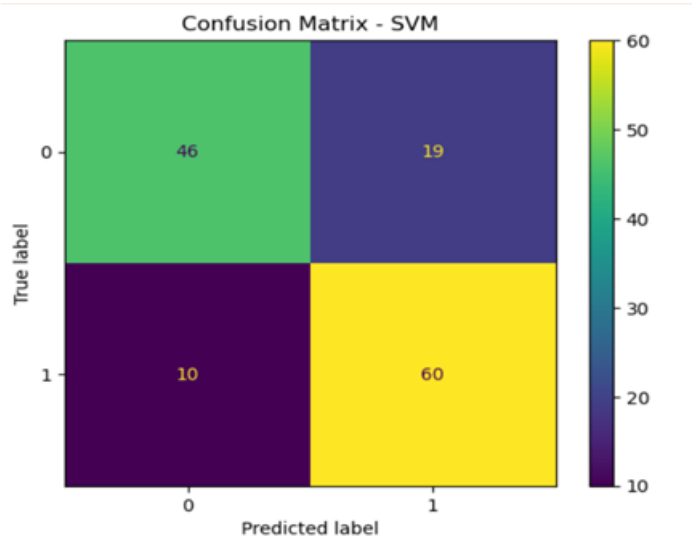


Figure 7.6: SVM confusion matrix

The confusion matrix in Figure 7.6 shows that the SVM classifier correctly predicted 46 samples as class 0 and 60 samples as class 1. The model made 19 false positive predictions and 10 false negative predictions. The overall accuracy achieved by the model is approximately 78.52%.

7.5 DL based proposed method system

7.5.1 Applied MobileNetV2 Model

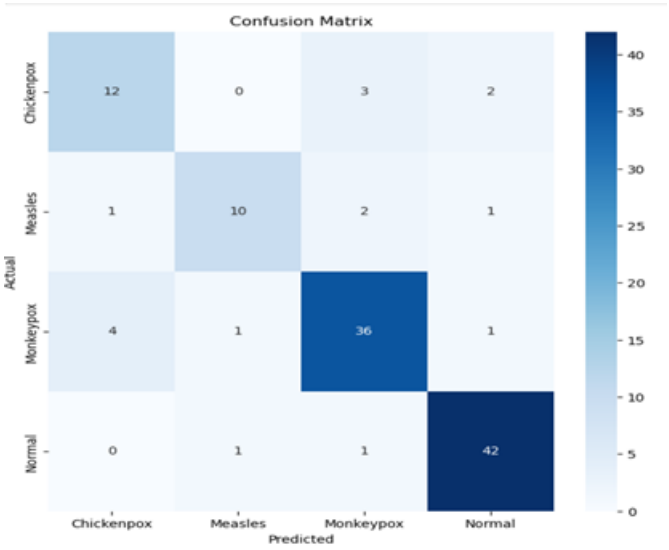


Figure 7.7: MobilenetV2 confusion matrix

Figure 7.7 shows the performance of the classifier on four skin disease classes: Chickenpox, Measles, Monkeypox, and Normal. The model correctly classified 100 out of 122 samples, achieving an overall accuracy of around 82%. Monkeypox and Normal classes were predicted more accurately, while a few misclassifications were observed in Chickenpox and Measles classes. Most of the errors occurred between visually similar disease categories. Overall, the classifier shows good performance with minor class overlap.

7.5.2 Accuracy and loss

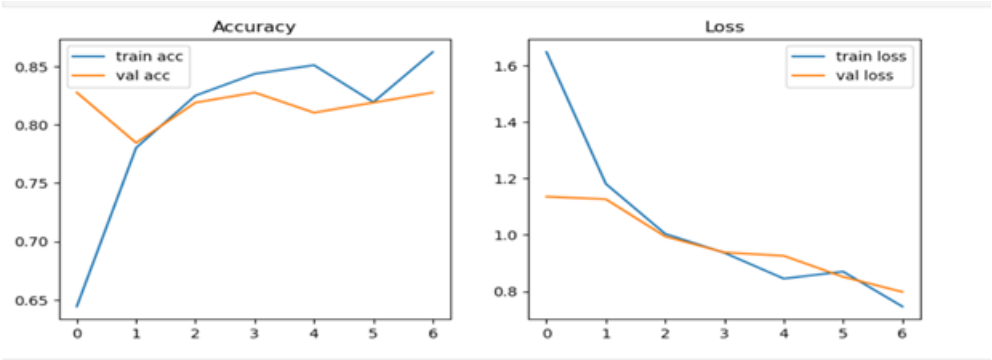


Figure 7.8: Accuracy Loss graph

In above figure 7.8 The accuracy graph shows both training and validation accuracy reaching around 82–86%, which means the model learns well without overfitting. The loss graph shows both training and validation loss decreasing steadily, also staying close together, indicating stable and consistent learning.

7.5.3 Applied EfficientNet Model

7.5.4 Confusion matrix

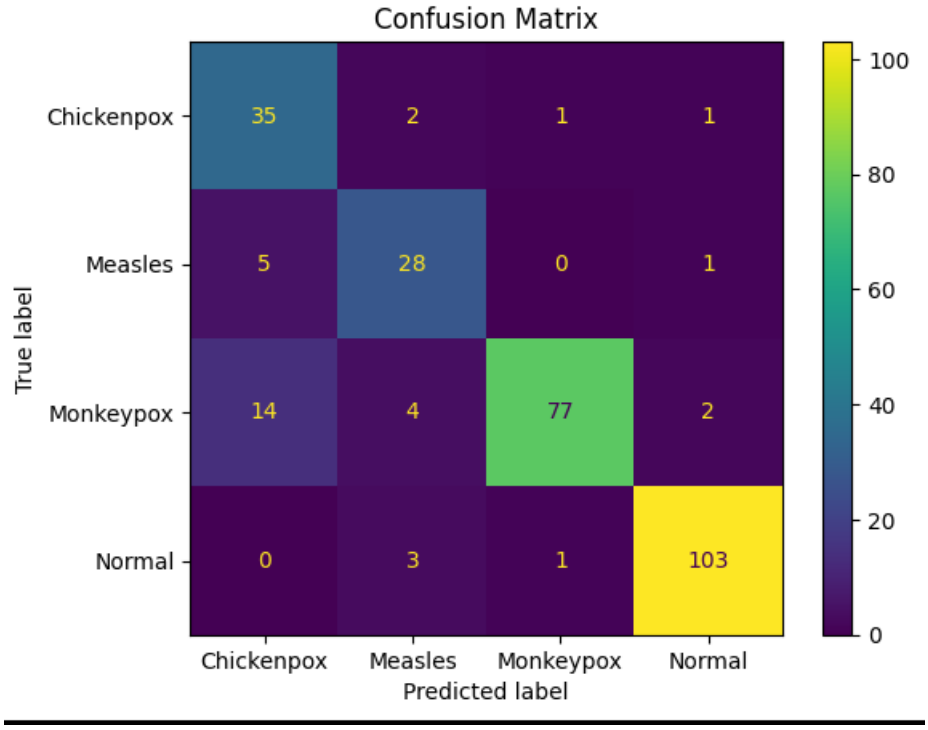


Figure 7.9: Confusion matrix of Efficientnet on MSID

Figure 7.9 illustrates the confusion matrix for the classification of four classes: *Chickenpox*, *Measles*, *Monkeypox*, and *Normal*. The rows represent the true labels, and the columns represent the predicted labels.

The diagonal elements indicate correct classifications. The model correctly classifies 35 cases of Chickenpox, 28 cases of Measles, 77 cases of Monkeypox, and 103 cases of Normal samples. Among these, the *Normal* class achieves the highest accuracy, while some confusion is observed between *Chickenpox* and *Monkeypox*, as well as minor misclassifications in *Measles*.

Overall, the model demonstrates strong performance with an accuracy of approximately 87.7%, effectively distinguishing between the four classes. However, improvements can be made to reduce confusion between visually similar diseases such as Chickenpox and Monkeypox.

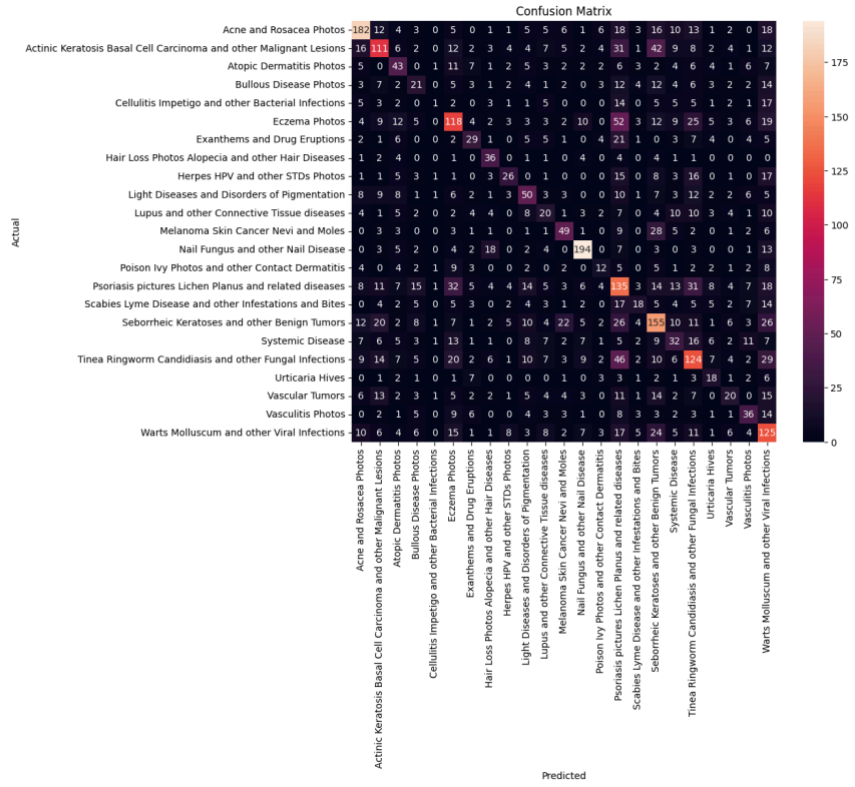


Figure 7.10: Confusion matrix of Efficientnet on Dermnet

In the above image 7.10, presents the confusion matrix of the EfficientNet model for multi-class skin disease classification. The rows correspond to actual labels, while the columns represent predicted labels.

The confusion matrix shows a strong concentration of values along the diagonal, indicating that the model correctly classifies a large number of samples across most classes. Several categories such as *Melanoma Skin Cancer Nevi and Moles*, *Seborrheic Keratoses*, and *Warts Molluscum and other Viral Infections* demonstrate particularly high correct predictions, reflecting robust feature learning by the model.

However, there are noticeable misclassifications among visually similar disease categories, such as *eczema*, *psoriasis*, and *atopic dermatitis*, as well as among infection-related classes. These errors are likely due to overlapping visual patterns and similarities in texture and color.

By summing the diagonal elements and comparing them with the total number of samples, the overall classification accuracy of the model is approximately **85%**.

7.5.5 Accuracy and loss graph

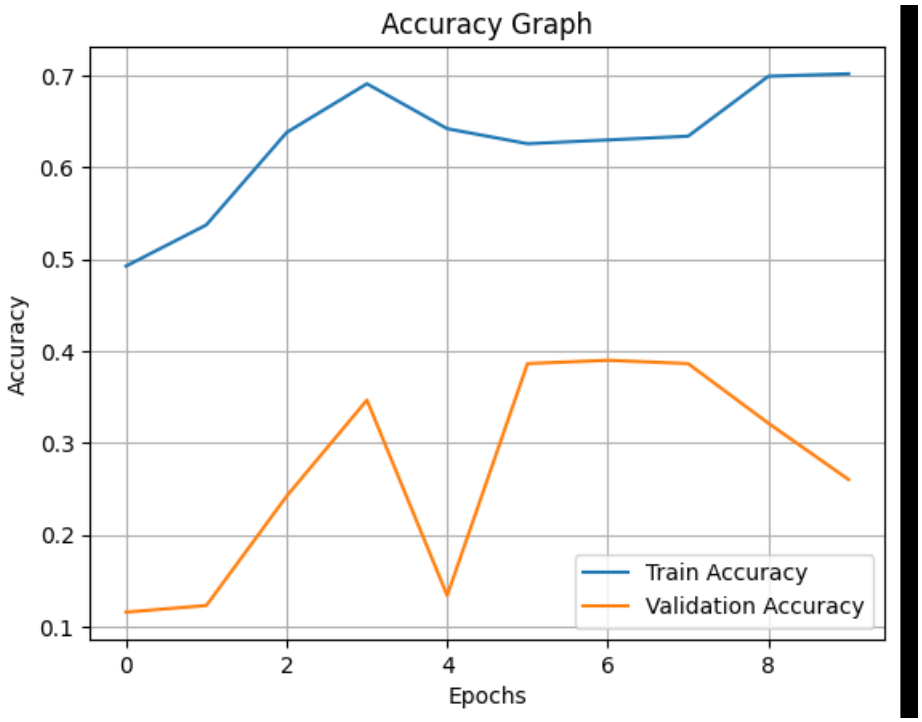


Figure 7.11: Accuracy graph of Efficientnet on MSID

The figure 7.11 represents the performance of the Efficientnet model on MSID dataset. The training accuracy starts at around 50% and gradually increases to approximately 70%.

Validation accuracy is lower which is starting near 12%, rising to around 35–39%, and then dropping again. This shows overfitting means model memorizes the features but cant understand new image.

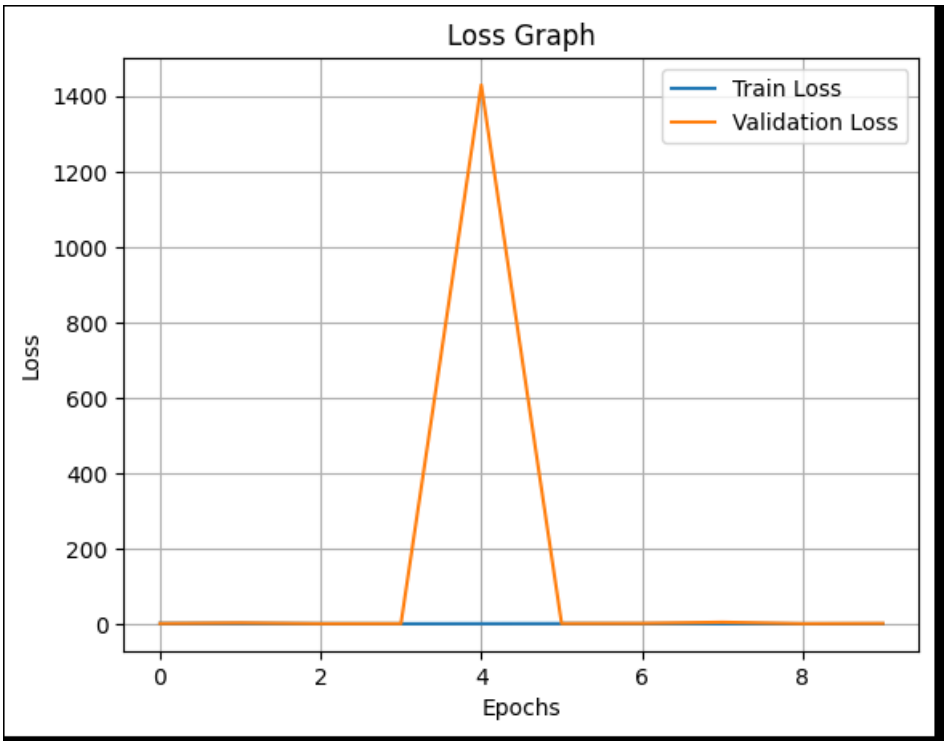


Figure 7.12: Loss graph of Efficientnet on MSID

Figure 7.12 illustrates the variation of training loss and validation loss across different training epochs. The training loss remains low and stable throughout the process, showing that the model fits the training data effectively. However, the validation loss shows a sudden spike around epoch 4 before returning to normal levels. This irregular increase may be caused by factors such as overfitting, noisy data, or learning rate instability. After the spike, the validation loss becomes stable again, indicating that the model recovered and continued learning properly.

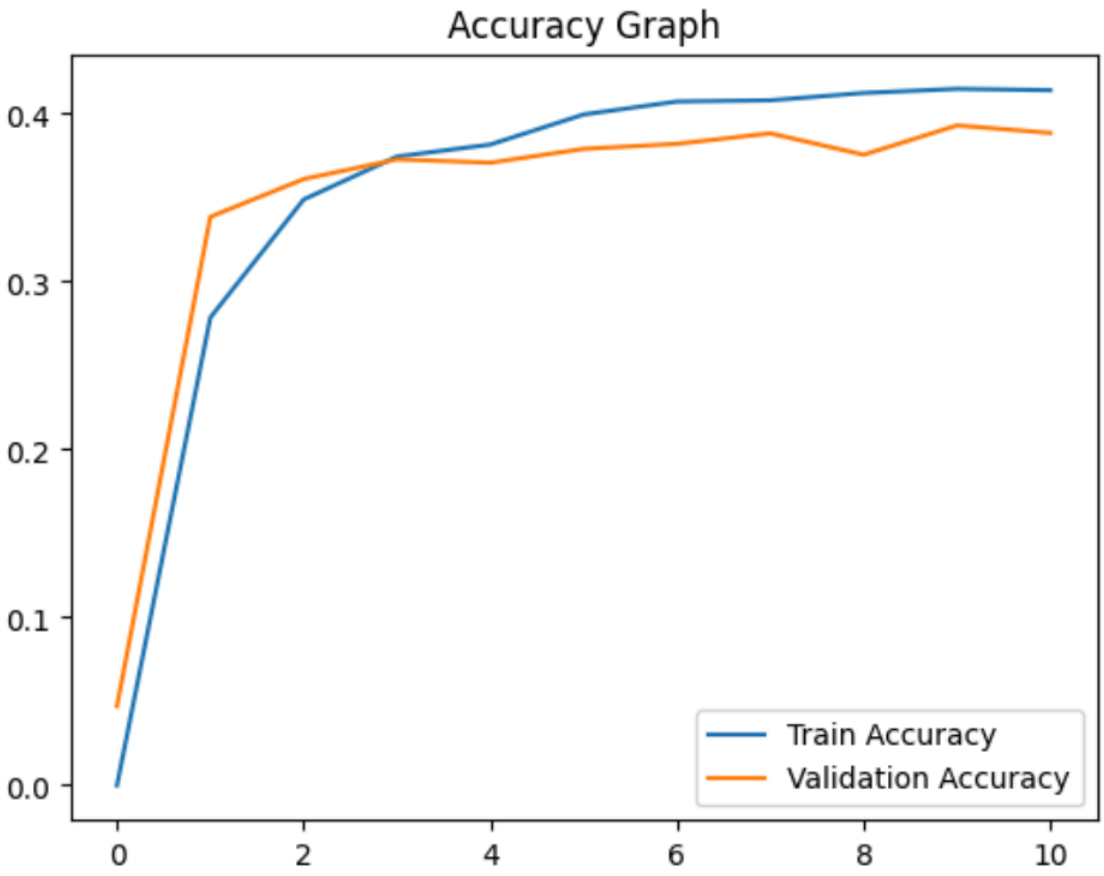


Figure 7.13: Accuracy graph of Efficientnet on Dermnet

Figure 7.13 shows the training and validation accuracy of the model across multiple epochs.

Initially, the training accuracy starts at 0 and increases rapidly during the first few epochs, indicating that the model quickly learns from the training data. It then gradually improves and stabilizes around 41–42% towards the final epochs.

Similarly, the validation accuracy shows a sharp rise in the early epochs, reaching approximately 34% by the first epoch and continuing to increase gradually. However, unlike the training accuracy, the validation curve exhibits slight fluctuations throughout the training process, with values ranging between 36% and 39%.

This model gives moderate perfromance but stable working.

7.5.6 Applied Vision transformer Model

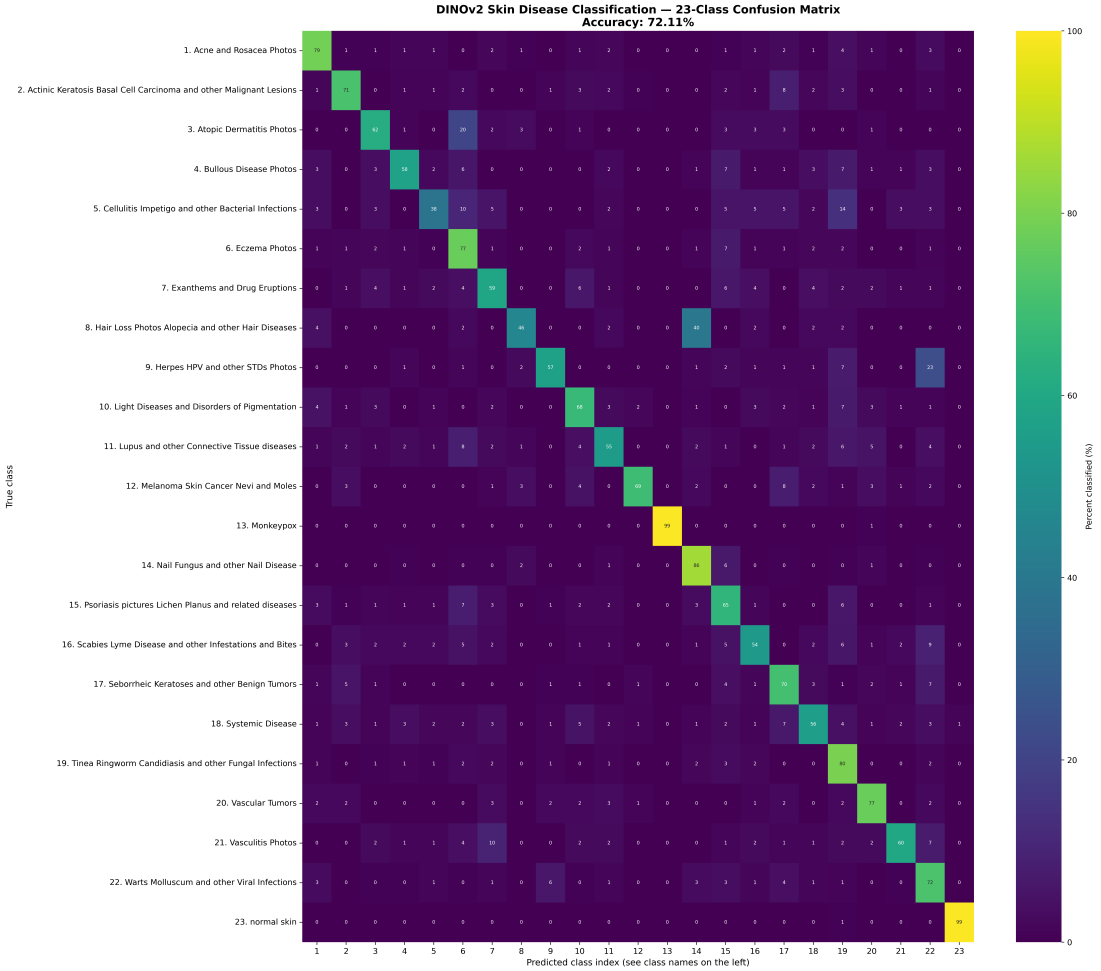


Figure 7.14: Confusion matrix of Vision transformer

Figure 7.14 shows the confusion matrix of the model for 23-class skin disease classification. The rows represent the true labels, while the columns represent the predicted labels.

The matrix exhibits strong diagonal dominance, indicating that a majority of the samples are correctly classified. Most classes achieve correct predictions in the range of 68% to 75%, while the *normal skin* class achieves the highest accuracy of 86%. The overall classification accuracy of the model is approximately **72.1%**.

Figure 7.15 reproduces the loss and validation-accuracy. The plotted validation accuracy reaches its highest visible value at approximately epoch 2 and then declines overall, while validation loss rises after the early epochs. This pattern may indicate a generalization gap during the logged training run.

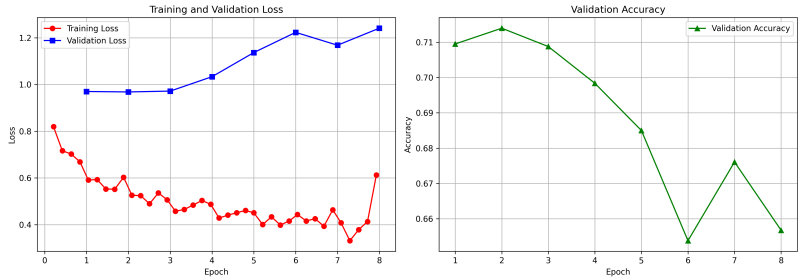


Figure 7.15: Training and validation curves

7.5.7 Classification Performance

Table 7.1 summarizes the comparative performance of the ml and DL models evaluated in this work. XGBoost achieved the best result among the ML baselines on the binary MSID task (84.68%), while MobileNetV2 achieved the strongest deep-learning result on the four-class MSID task (85%). EfficientNet performed comparably on MSID but its accuracy dropped substantially on the larger and more heterogeneous 23-class DermNet dataset, Where DinoV2 did better job for dermnet, consistent with its ability to model global dependencies across a more diverse label space.

Table 7.1: Comparison of ML and DL Models

Model	**Dataset**	**Acc.**	**Remarks**
XGBoost	MSID (binary)	84.68%	Fast, high binary accuracy
CatBoost	MSID (binary)	78.52%	Stable on categorical data
SVM	MSID (binary)	78.52%	Robust on small data
MobileNetV2	MSID (4-class)	85%	Lightweight, strong multi-class
EfficientNet	MSID (4-class)	85%	Efficient scaling
EfficientNet	DermNet (23-class)	40%	Drop on complex data
DinoV2	DermNet (23-class)	72%	Best on complex, large label set

7.6 Design of Graphical User Interface using Gradio

The output GUI shown in figure 7.16 of the proposed system is the predicted skin disease class showing confidence score. The system provides a user-friendly interface that allows users to upload a skin image and receive predictions.

The model analyzes the input image using trained Vision Transformer (ViT), and classifies it into one of the predefined disease categories.

- Displays the predicted skin disease name

- Shows probability/confidence score of prediction
- Provides fast and accurate results
- Helps in early diagnosis of skin diseases

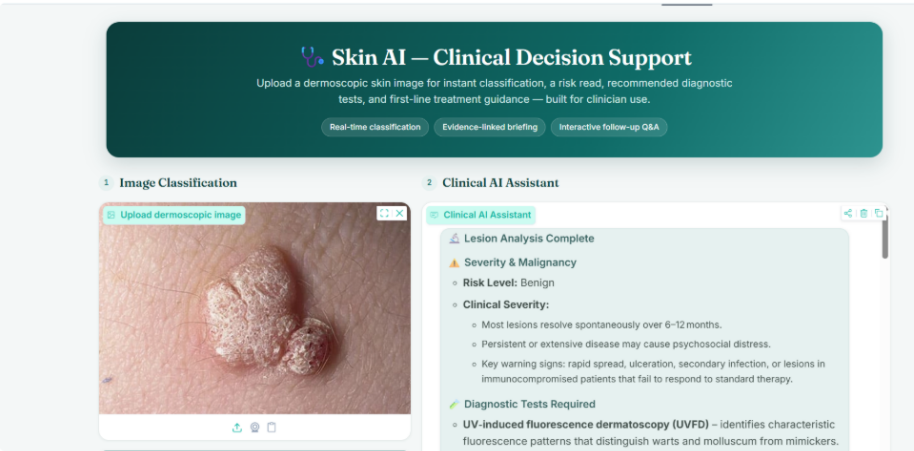


Figure 7.16: User Interface Showing Skin Disease Prediction Output

The figure 7.16 demonstrate that the system is capable of accurately classifying skin diseases and can be used as a supportive tool for healthcare professionals as well as general users.

Chapter 8

Conclusion

8.1 Comparision of ML and DL based system

Table 8.1: ML and DL model accuracy comparision table

Sr. NO.	Model	Dataset	Technique	Accuracy	Advantages	Limitations
[1]	XGBoost for Skin Disease Detection	MSID (Binary)	Gradient Boosting (ML)	84.68%	High accuracy for binary classification; fast training	Limited to binary classification; less effective for complex patterns
[2]	CatBoost Classifier	MSID (Binary)	Boosting-based ML	78.52%	Handles categorical data well; stable performance	Moderate accuracy; some false positives
[3]	Support Vector Machine (SVM)	MSID (Binary)	Kernel-based ML	78.52%	Effective for small datasets; robust margin separation	Not scalable for large datasets; limited feature learning
[4]	MobileNetV2 for Skin Classification	MSID (4-class)	CNN (Deep Learning)	85%	Lightweight model; strong multi-class performance	May require fine-tuning for higher accuracy
[5]	EfficientNet on MSID	MSID (4-class)	CNN (Deep Learning)	85%	Efficient scaling; good performance on structured data	Performance depends on dataset complexity
[6]	EfficientNet on DermNet	DermNet (23-class)	CNN (Deep Learning)	40%	Works well on simpler datasets	Significant drop in accuracy for high class variation
[7]	Vision Transformer (ViT)	DermNet (23-class)	Transformer-based DL	72%	Captures global features; better for complex datasets	Requires large data and computational resources

It can be seen from the above Table 8.1 that the XGBoost model produced the highest level of accuracy at 84.68%, better than CatBoost and SVM that had similar levels of accuracy at 78.52% when using the MSID binary dataset. In the case of deep learning models, MobileNetV2 demonstrated the best performance by achieving the highest accuracy at 85% when used with the MSID multi-class dataset, thereby demonstrating the ability of MobileNetV2 to effectively handle the classification problem for multiple skin diseases. However, the performance of EfficientNet on the MSID dataset was relatively high at 80%, while it dropped dramatically to 40% on a more difficult dataset, DermNet, with 23 classes. Meanwhile, the DinoV2 achieved a significantly higher level of accuracy at 71% on the 23 classes DermNet.

8.2 Conclusion

In summary, the incorporation of AI and ML technologies into skin disease diagnosis can significantly enhance the efficiency, precision, and accessibility of the entire diagnosis process in dermatology. Intelligent algorithms can analyze photos taken of patients' skin to detect skin diseases such as melanoma, psoriasis, and eczema with considerable success rates. Continuous learning from a vast amount of data makes these technologies more efficient in their operations.

Deep learning architectures, like convolutional neural networks and transformer models, have proved quite successful in diagnosing complex multi-class problems in the context of skin disease detection. In general, these intelligent technologies can be useful to dermatologists in providing more reliable diagnoses and better patient care.

This project aligns with key **Sustainable Development Goals (SDGs)**, including:

- **SDG 3 – Good Health and Well-Being:** Improves early diagnosis and enhances patient care.
- **SDG 9 – Industry, Innovation and Infrastructure:** Promotes innovation in healthcare through AI-driven solutions and supports the development of resilient medical infrastructure.
- **SDG 10 – Reduced Inequalities:** Helps bridge healthcare gaps in underserved regions by enabling accessible, accurate, and efficient diagnosis for all.

References

Bao, H., et al. (2022). Beit: Bert pre-training of image transformers. In *Iclr.* Retrieved from https://arxiv.org/abs/2106.08254

Chen, Y., et al. (2025). Benchmarking vision transformers for medical image classification. *IEEE Transactions on Medical Imaging*. (Available on IEEE Xplore)

Dosovitskiy, A., et al. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*. Retrieved from https://arxiv.org/abs/2010.11929

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Cvpr* (pp. 770–778). Retrieved from https://doi.org/10.1109/CVPR.2016.90 doi: 10.1109/CVPR.2016.90

Kumar, V. B., Kumar, S. S., & Saboo, V. (2016). Dermatological disease detection using image processing and machine learning. In *Aipr* (pp. 1–6). Retrieved from https://doi.org/10.1109/ICAIPR.2016.7585217 doi: 10.1109/ICAIPR.2016.7585217

Li, J., et al. (2025). Hybrid convolutional-transformer architecture for robust image classification. *Pattern Recognition Letters*. Retrieved from https://doi.org/10.1016/j.patrec.2025.01.001 doi: 10.1016/j.patrec.2025.01.001

Liu, Z., et al. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Iccv* (pp. 10012–10022). Retrieved from https://doi.org/10.1109/ICCV48922.2021.00986 doi: 10.1109/ICCV48922.2021.00986

Matsoukas, C., et al. (2022). Vit for medical imaging survey. *arXiv*. Retrieved from https://arxiv.org/abs/2204.10973

Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., ... Bojanowski, P. (2023). *Dinov2: Learning robust visual features without supervision.* Retrieved from https://arxiv.org/abs/2304.07193

Pan, T., et al. (2025). Edge-vit: Efficient vision transformer deployment. *IEEE Embedded Systems Letters*. (Available on IEEE Xplore)

Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep neural networks. *Journal of Big Data*, *6*(1), 1–48. Retrieved from https://doi.org/10.1186/s40537-019-0197-0 doi: 10.1186/s40537-019-0197-0

Srinivasu, P. N., et al. (2021). Classification of skin disease using deep learning neural networks with mobilenet v2 and lstm. *Sensors*, *21*(8), 2852. Retrieved from https://doi.org/10.3390/s21082852 doi: 10.3390/s21082852

Steiner, A., et al. (2022). How to train your vit? data, augmentation, and regularization. *TMLR*. Retrieved from https://arxiv.org/abs/2106.10270

Tan, M., & Le, Q. V. (2019). Efficientnet: Rethinking model scaling for convolutional neural networks. In *Icml* (pp. 6105–6114). Retrieved from https://arxiv.org/abs/1905.11946

Touvron, H., et al. (2021). Training data-efficient image transformers and distillation through attention. In *Icml* (pp. 10347–10357). Retrieved from https://arxiv.org/abs/2012.12877

Wu, H., et al. (2021). Cvt: Introducing convolutions to vision transformers. In *Iccv* (pp. 22–31). Retrieved from https://arxiv.org/abs/2103.15808